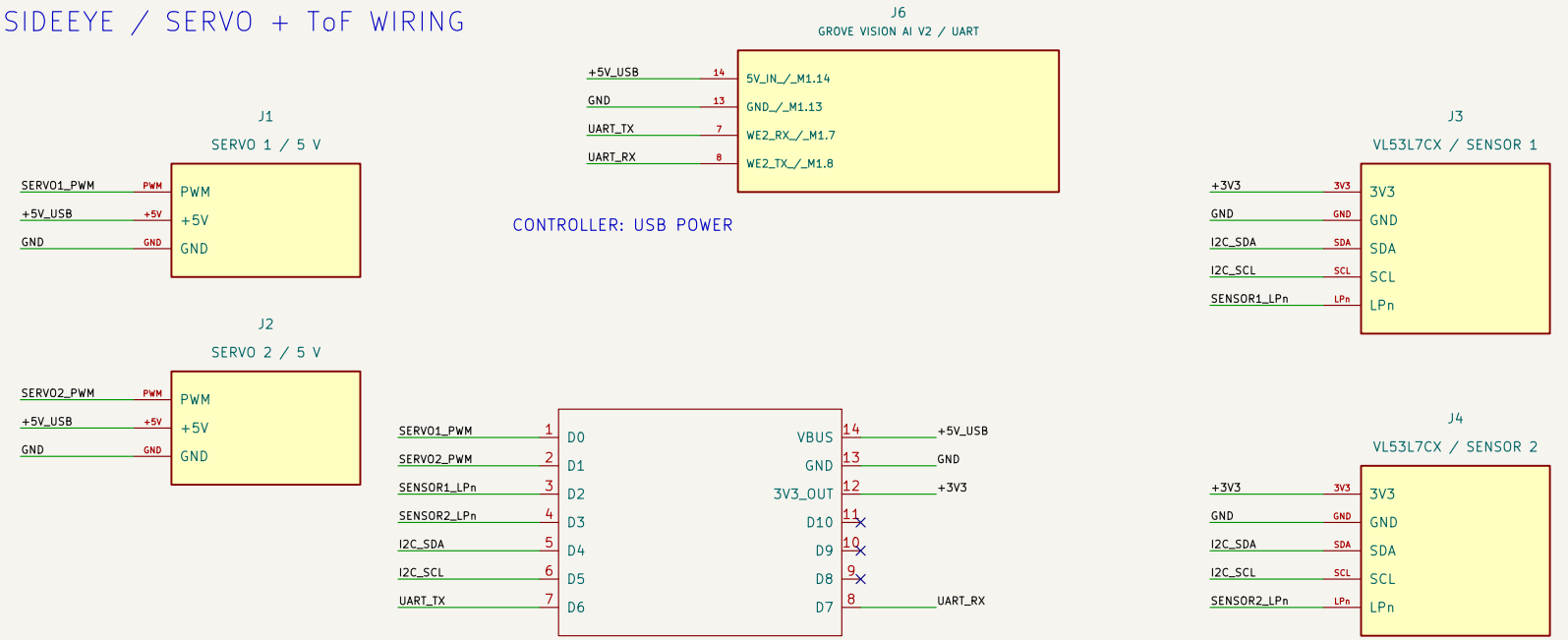
SIDEEYE / SERVO + ToF WIRING



USB-C ON U1 = ONLY POWER INPUT
5V USB -> U1 pin 14 -> servos + Grove
3V3 regulator -> U1 pin 12 -> ToF x2

D0 / GPIO1 = servo 1 PWM
D1 / GPIO2 = servo 2 PWM
D2 / GPIO3 = sensor 1 LPn
D3 / GPIO4 = sensor 2 LPn
D4 / GPIO5 = shared SDA
D5 / GPIO6 = shared SCL
D6 / GPIO43 = UART TX to WE2
D7 / GPIO44 = UART RX from WE2

WIRING NOTES

1. U1 = XIAO ESP32-S3. J6 = Grove Vision AI V2 socket interface to Himax WiseEye2 HX6538.
2. Fit a selective interposer / wire adapter: pass ONLY M1 pins 7, 8, 13, 14 to Grove.
   Isolate M1 pins 1-6 and 9-12, especially D3 (Grove RESET) and 3V3 (separate regulators).
   D0-D5 connect to peripherals on the XIAO side. This drawing requires that isolation.
3. Sole power input: U1 USB-C. U1 pin 14 supplies +5V_USB to both servos and Grove M1.14.
   U1 pin 12 supplies +3V3 to both ToF modules. All grounds common; PWM/UART logic is 3.3 V.
   Verify USB source, cable and XIAO VBUS path for BOTH servo stall currents plus all electronics.
   Check 3V3 regulator headroom for both sensors; use short servo power returns and local decoupling.
4. D4/D5 form one shared I2C bus for J3/J4. Ensure SDA/SCL pullups to 3V3 on the modules;
   account for parallel pullups. Start at 400 kHz. J1-J4 show signal names, not connector pin order.
5. At boot drive both LPn low; raise sensor 1 LPn, set address to 0x2A, then raise sensor 2 (0x29).
   Addresses above are 7-bit; ST API may use 0x54 / 0x52. Repeat after power loss.
6. Use 3.3 V-compatible VL53L7CX modules with I2C_RST held low and INT unused (polling).
   D6 TX -> WE2 PB6 RX; D7 RX <- WE2 PB7 TX. Select UART in the Grove firmware/host driver.

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